Trøndelagkonkurranse 2026 - Level II

Second Round - October 24, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Nils draws two different parallel lines A_1 and A_2 on the plane. The lines B and C each intersect both A_1 and A_2. Suppose that a line L satisfies the following conditions:

- L intersects A_1 at point P.
- L intersects A_2 at point S.
- L intersects B and C at the points Q and R.
- The order of the points P, Q, R and S in L is $P - Q - R - S$.

Show that Bianca can choose L such that the lengths of the segments PQ, QR and CD are all equal.

Problem 2

Let n be a positive integer. Each of the numbers $1, 2, 3, \ldots, n$ is written on a blackboard once. At each step, we can erase two numbers on the blackboard, say x and y, and replace them with the single number $|x - y|$. After several steps, there is only one number on the blackboard. For what values of n can this number be equal to 0?